

Semester Project: Domain Discovery, Recommendation, and Graph Intelligence

Milestone: Prototype-Based Clustering - K-Means Experiments (Week 7)

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1. Prior Milestones Summary

Week	Milestone	Key output used here
3	Dataset Charter & EDA	steam_v1.parquet: 480,025 records. Star schema on AppID.
5	Feature Engineering	X_numeric (87k x 26), X_text (87k x 5k). PCA/SVD models in models.pkl.
7	Clustering (this)	Tags-only binary matrix. K=6 cluster labels -> artifacts/clustering/

2. Clustering Setup

- Representation: tags-only binary matrix from genres + categories + tags columns.
- Vocabulary filter: min_freq=30 -> 5,397 tag dimensions retained.
- Input universe: 69,943 games with non-empty tag fields.
- Rationale: tag tokens are community-curated labels (e.g. 'roguelike', 'local co-op') with less noise than free-text.

3. Lloyd's Algorithm - Manual Descent Verification

To verify sklearn's K-means implements Lloyd's algorithm correctly, one iteration (assignment + centroid update) is run manually from 5 random seeds on PCA-10D numeric features.

Seed	J_before	J_after	dJ	Descent?
1	890,570	665,135	-225,435	Yes
2	1,062,161	898,191	-163,970	Yes
3	1,114,696	663,283	-451,412	Yes
4	1,151,948	882,097	-269,850	Yes
5	1,038,707	718,923	-319,784	Yes

All 5 seeds confirm strict $J_{\text{after}} \leq J_{\text{before}}$, verifying the implementation.

4. Ablation Study - Representation Variants

Variant	Silhouette	Largest cluster (%)
Raw sparse (binary tags)	0.026	24.9%
SVD only (50D dense)	0.064	21.7%
SVD + L2 (chosen)	0.077	22.0%

- Raw sparse: K-means with Euclidean distance on 5,397D binary vectors -> lowest quality.
- SVD 50D: dimensionality reduction concentrates variance, silhouette improves substantially.
- SVD + L2 (chosen): L2 normalization maps to unit hypersphere, converting distance to cosine. Best result.

5. K-Sweep - Choosing Number of Clusters

K	Inertia	Silhouette	Min cluster	Max cluster
4	31,128	0.0796	13,592	21,442

29,175

0.0767

4,773

15,353

27,664

0.0836

958

12,314

10

26,357

0.0946

958

11,747

K=6 selected: K=8/10 produce smaller minimum clusters (958 games, ~1.4%) impractical for segmentation. K=6 balances compactness and semantic interpretability.

6. Final Clustering - K=6 Cluster Profiles

Silhouette: 0.0767 | Inertia: 29,175 | Top tags by lift = $P(\text{tag}|\text{cluster}) / P(\text{tag})$

- C0 (4,773 games | 100%): Creative/Dev Tools -> video production: 31 (14.65x), audio production: 31 (14.65x), photo editing: 21 (14.65x), animation & modeling: 21 (14.47x), photo editing (14.41x)
- C1 (15,353 games | 322%): Action -> combat: 15 (3.17x), action-adventure: 138 (3.12x), side scroller: 29 (3.08x), action: 316 (3.07x), action: 64 (3.05x)
- C2 (13,576 games | 284%): Narrative/Adventure -> text-based: 6 (3.99x), text-based: 8 (3.95x), steam cloud (3.59x), story rich: 71 (3.06x), great soundtrack: 33 (2.85x)
- C3 (14,060 games | 295%): Strategy/Puzzle -> card game: 11 (3.77x), logic: 28 (3.7x), logic: 19 (3.63x), building: 24 (3.27x), adventure: 11 (3.24x)
- C4 (12,112 games | 254%): Local Multiplayer -> local co-op: 12 (5.77x), 4 player local: 13 (5.77x), 4 player local: 12 (5.77x), shared/split screen co-op (5.75x), shared/split screen pvp (5.74x)
- C5 (10,069 games | 211%): Casual/Hidden Object -> hidden object: 6 (6.45x), hidden object: 5 (5.97x), hidden object: 7 (5.89x), point & click: 14 (5.52x), point & click: 12 (5.45x)

7. ARI Stability Across Seeds

Mean off-diagonal ARI: 0.4902 | Min pairwise ARI: 0.3423

ARI ~0.49 indicates moderate-to-good agreement: broad segments (Action, Strategy, Local MP) are consistently recovered. Boundary games fluctuate between similar clusters.

8. Reproducible Pipeline

```
python src/ingest.py           # Step 1
python src/feature_engineering.py # Step 2
python src/clustering_week7.py  # Step 3 - this milestone
```

All outputs deterministic (random_state=42).

9. Ethics and Access Note

- Clustering operates on game catalog metadata only. No user data involved.
- CC0 license; unrestricted academic use.
- Tag vocabulary reflects Steam community labels; English/AAA titles dominate by frequency.